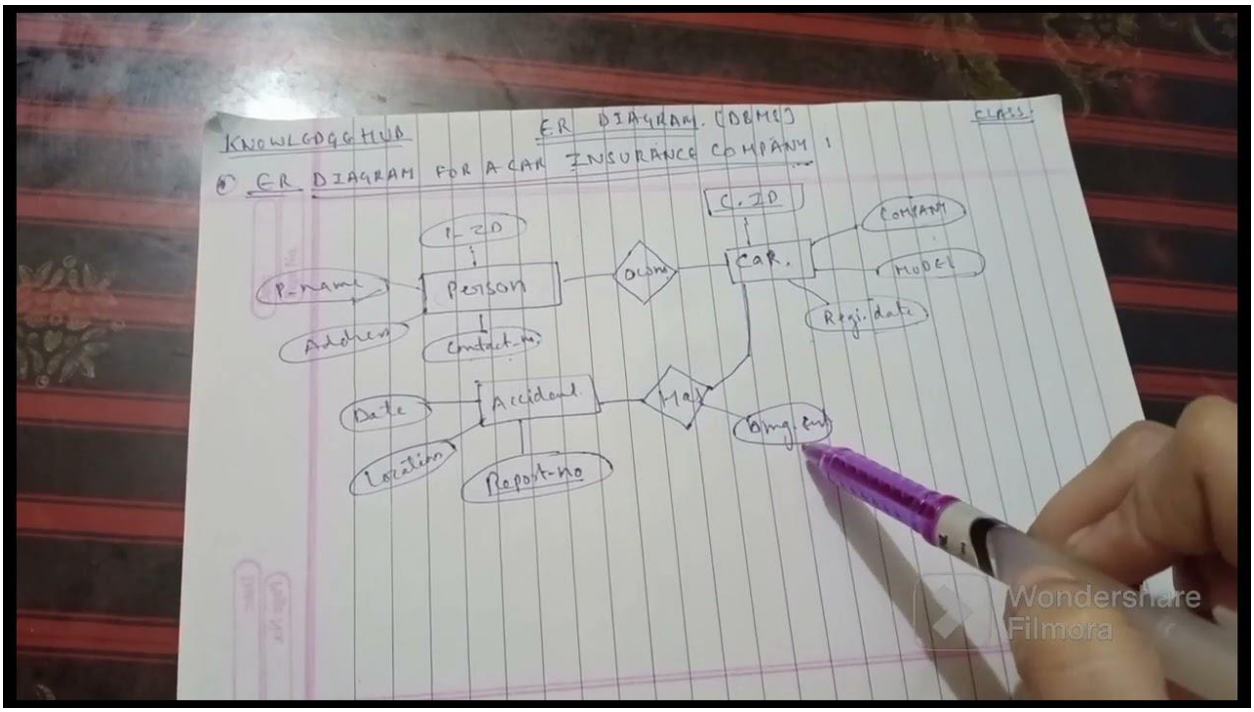
1.Draw the er diagram for car insurance company.



2.Explain DDL,DML,Commands with SQL query
ans-

DDL (Data Definition Language)

- **Purpose:** Defines and manages database schema and structure.
- **Commands:**
  - **CREATE**: Creates database objects (tables, views).
    - *Example:* `CREATE TABLE Employees (empid INT, empname VARCHAR(100), empsalary DECIMAL(10,2));`
  - **ALTER**: Modifies existing database objects.
    - *Example:* `ALTER TABLE Employees ADD department_no INT;`
  - **DROP**: Deletes database objects.
    - *Example:* `DROP TABLE Employees;`

DML (Data Manipulation Language)

- **Purpose:** Manages and manipulates data within the database.
- **Commands:**
  - **SELECT**: Retrieves data from database tables.
    - *Example:* `SELECT * FROM Employees;`

- **INSERT**: Adds new data to tables.
  - *Example:* `INSERT INTO Employees (empid, empname, empsalary) VALUES (101, 'John Doe', 25000);`
- **UPDATE**: Modifies existing data in tables.
  - *Example:* `UPDATE Employees SET empsalary = 27000 WHERE empid = 101;`
- **DELETE**: Removes data from tables.
  - *Example:* `DELETE FROM Employees WHERE empid = 101;`

These commands are essential for creating, managing, and manipulating database structures and data.

3.Department (department no.,department name,department location):
Employees (empid,empname,empsalary);
q1. find all the employee who have salary greater than 20000
q2.display the rows whoes salary ranges from15000 to 30000.
q3. calculate the total and avg. salary amount of emp table.
q4. display department no. from the table employee avoiding the duplicate values.
q5. select name of department whoes empid=103

ANS-

a-

```
SELECT *

FROM Employees

WHERE empsalary > 20000;
```

b-

```
SELECT *

FROM Employees

WHERE empsalary BETWEEN 15000 AND 30000;
```

c-

```
SELECT

   SUM(empsalary) AS total_salary,
```

AVG(empsalary) AS average_salary

FROM Employees;

d-

SELECT DISTINCT department_no

FROM Employees;

e-

SELECT d.department_name

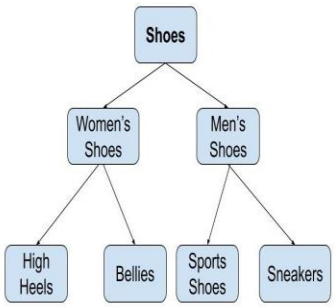
FROM Employees e

JOIN Department d ON e.department_no = d.department_no

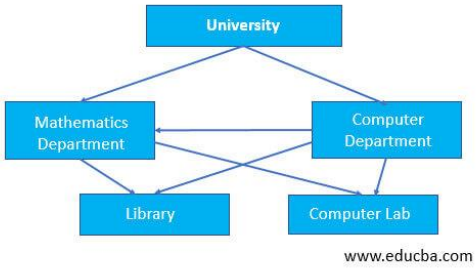
WHERE e.empid = 103;

Q6- describe data models and it types

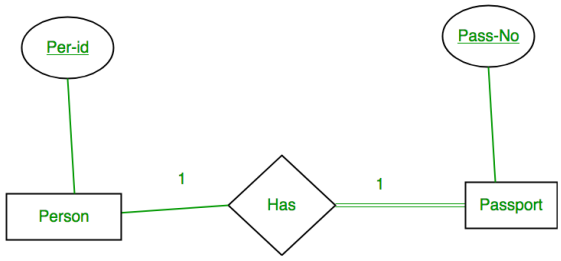
Ans-

1. **Hierarchical Model**:
    - **Structure:** Data is organized in a tree-like structure with parent-child relationships.
    - **Example:** File systems.
    - 

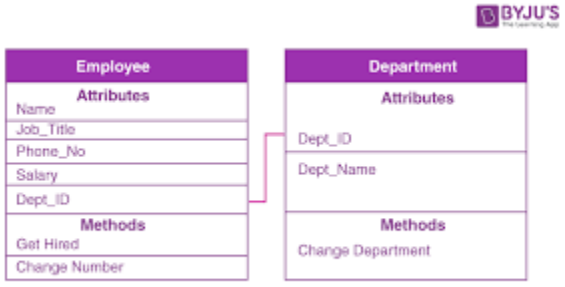
Hierarchical Model
2. **Network Model**:
    - **Structure:** Data is organized as a graph with nodes and connections, allowing multiple relationships between entities.
    - **Example:** Network databases.



3. **Relational Model**:
    - **Structure:** Data is organized into tables (relations) with rows and columns, where tables can be linked by keys.
    - **Example:** SQL databases (e.g., MySQL, PostgreSQL).

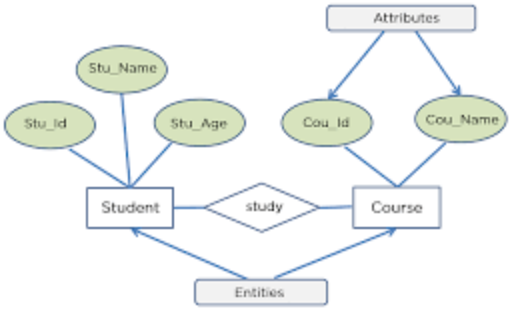


4. **Object-Oriented Model**:
    - **Structure:** Data is represented as objects, similar to objects in object-oriented programming.
    - **Example:** Object-oriented databases (e.g., db4o).



Object Oriented Model

5. **Entity-Relationship Model**:
    - **Structure:** Data is represented using entities (objects) and relationships between them, often visualized in ER diagrams.
    - **Example:** Conceptual database design.

- 

Each model has its own strengths and is suited to different types of applications and data management needs.

Q7 -define the following term
superkey,candidate key,primary key,foreign key

ANS -

Superkey:

- **Definition:** A superkey is a set of one or more columns (attributes) that uniquely identifies a row in a table.
- **Advantage:** Ensures that every row in a table can be uniquely identified.

Candidate Key:

- **Definition:** A candidate key is a minimal superkey, meaning it is a superkey with no unnecessary attributes; it uniquely identifies a row.
- **Advantage:** Provides the smallest possible set of columns needed to uniquely identify a row, reducing redundancy.

Primary Key:

- **Definition:** A primary key is a candidate key selected to uniquely identify rows in a table. It cannot contain `NULL` values.
- **Advantage:** Ensures the uniqueness and integrity of each row and is used to establish relationships with other tables.

Foreign Key:

- **Definition:** A foreign key is a column or set of columns in one table that - to the primary key of another table.

- **Advantage:** Establishes and enforces a link between tables, ensuring referential integrity and consistency across the database.

Q8.consider the following schema for hotel data base
hotel (hotel no,hotelname,city)
room (room no.,hotel no.,type,price)
a.list all the hotels
b.list the price and the type of all rooms at the Grosvenor hotel
c.find the avg ,min.,max,room rent for the rooms of hotel in the database.

ANS -

Schema :

hotel (hotel_no, hotel_name, city)

room (room_no, hotel_no, type, price)

a. List all the hotels:

```
SELECT * FROM hotel;
```

b. List the price and the type of all rooms at the Grosvenor hotel:

```
SELECT r.price, r.type
FROM room r
JOIN hotel h ON r.hotel_no = h.hotel_no
WHERE h.hotel_name = 'Grosvenor';
```

c. Find the average, minimum, and maximum room rent for the rooms in the database:

```
SELECT
    AVG(price) AS average_price,
    MIN(price) AS minimum_price,
    MAX(price) AS maximum_price
FROM room;
```

Q9 -What is er diagram draw ER diagram for hospital management

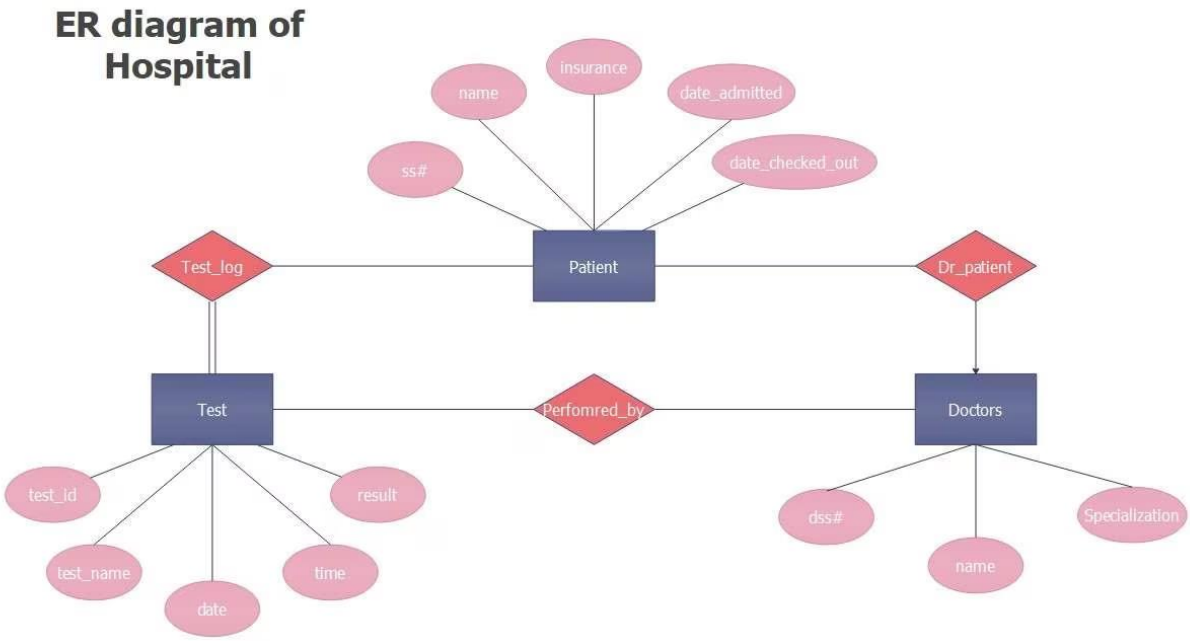
ANS -

An Entity-Relationship (ER) diagram is a visual tool used in database design. It shows how different pieces of data (entities) are related to each other.

It helps in designing the database by mapping out how different pieces of information relate to each other

Notations :

- **Entity**: Rectangle
- **Attribute**: Oval
- **Relationship**: Diamond
- **Primary Key**: Underlined Attribute
- **Foreign Key**: Dashed Underline or Line
- **Cardinality**: Symbols or Numbers



Q10- which tier architecture is most important will designing a database justify your answer

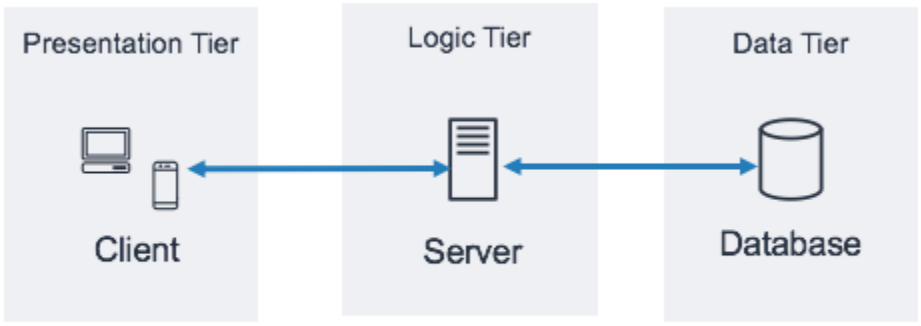
ANS -

In database management systems (DBMS), the **three-tier architecture** is often considered most important when designing a database. This architecture comprises:

1. **Presentation Tier**: User interface layer.
2. **Application Tier**: Business logic and application processes.
3. **Tier**: Data storage and management

This architecture helps in creating more modular, manageable, and scalable database systems.

In summary, the three-tier architecture provides a clear separation of concerns, improving scalability, security, and maintainability



Q11.list various ddl commonds. Explain with example

ANS -

1. **CREATE**: Creates new database objects (e.g., tables, views).
   Ex- CREATE TABLE employee (
   empno INT PRIMARY KEY,
   name VARCHAR(50),
   empcity VARCHAR(50)
   );

2. **ALTER**: Modifies existing database objects.
   Ex- ALTER TABLE employee
   ADD salary DECIMAL(10, 2);
   ALTER TABLE employee
   MODIFY empcity VARCHAR(100);

3. **DROP**: Deletes database objects permanently.
   Ex- DROP TABLE employee;

4. **TRUNCATE**: Deletes all records in a table.
   Ex- TRUNCATE TABLE employee;

5. **RENAME**: Renames database objects.
   Ex- RENAME TABLE employee TO staff;

6. **COMMENT**: Adds comments to database objects.

Ex- COMMENT ON TABLE employee IS 'This table stores employee details';
COMMENT ON COLUMN employee.salary IS 'The salary of the employee';

DDL commands are essential for setting up the structure of a database and managing its objects effectively.

Q12 -.employee (empno,name,empcity) company (empno,company name,salary)

a.write a SQL query to display employee name and company

. b.write a sql query to display employee name,employee city,company name and salary of all the employee whoes salary is less than 10000

c.write a query to display all the employee working in 'XYZ' company

d.find the employee with the lowest salary and display there name ,company name and salary.

ANS -

a) SELECT e.name, c.company_name

FROM employee e

JOIN company c ON e.empno = c.empno;

name	company_name
Alice	ABC Corp
Bob	XYZ Ltd
Charlie	ABC Corp

b) SELECT e.name, e.empcity, c.company_name, c.salary

FROM employee e

JOIN company c ON e.empno = c.empno

WHERE c.salary < 10000;

name	empcity	company_name	salary

Alice	New York	ABC Corp	9500
Bob	Chicago	XYZ Ltd	9000

c) SELECT e.name, e.empcity, c.company_name, c.salary

FROM employee e

JOIN company c ON e.empno = c.empno

WHERE c.company_name = 'XYZ';

name	empcity	company_name	salary
Bob	Chicago	XYZ Ltd	9000

d) SELECT e.name, c.company_name, c.salary

FROM employee e

JOIN company c ON e.empno = c.empno

WHERE c.salary = (SELECT MIN(salary) FROM company);

name	company_name	salary
Charlie	ABC Corp	8500